Measuring What We Mean
Construct Validity When Measures Are Cheap

A position paper on measurement selection in the age of AI

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Status. This document states a research program and the evidence required to support it. Empirical results on real survey and AI measures remain *intentionally empty*: they await the preregistered data and evaluation harness of Section 8, not missing engine primitives. The companion package `cvprofiles` 2.5.0 now ships the holdout workflows (restriction-stage tags and units-split) and the additive coverage uncertainty band under synthetic gates; those layers are illustrated in Section 7. What follows is the framework, its first-principles derivation, controlled synthetic illustrations, and a preregistered plan for the real-world proof of concept. Code: `SCA2_PofW`; engine: `cvprofiles` (tag `v2.5.0`).

Statement of significance

Scientists rarely observe constructs such as trust, ideology, culture, or risk preference directly. They work through measures. Artificial intelligence changes the economics of this problem: researchers can now generate many plausible measures quickly—prompted responses, embeddings, classifiers, synthetic agents, fine-tuned policies. When measures become cheap, choosing and validating them becomes the scarce scientific task.

This paper proposes a simple discipline. Define the construct; declare a menu of candidate measures; state observable validity tests implied by the construct; retain the measures that survive; and report which scientific conclusions are robust across the surviving set. The framework treats conventional survey measures and AI-generated measures symmetrically. It completes, operationally, a construct-validity program that recent econometric work formalized as partial identification over measurement menus and explicitly called for principled, widely accepted methods of construct validation [14]. The intended contribution is a *systems view* of empirical science under cheap measurement (Figure 2)—separating hypothesis generation, measurement, construct validation, and estimation—together with a general, auditable method for using construct-validity evidence to select among heterogeneous measures and propagate remaining measurement uncertainty into scientific conclusions. That method is implemented in open software, illustrated on controlled data-generating processes, and designed for a real-world cultural-preferences application once frozen inputs and the evaluation harness support the intended analysis.

Abstract

Artificial intelligence has sharply reduced the cost of producing candidate measures of latent constructs. A researcher can measure trust, sentiment, ideology, or cultural preference with a survey item, a composite score, a prompted language model, or a fine-tuned policy. The resulting abundance creates a selection problem: which measures warrant the interpretation researchers give them, and which empirical conclusions survive reasonable measurement choice? Measurement selection is formulated around four objects: a construct C, a declared menu of candidate measures $\mathscr{M}$, a set of theory-derived validity tests $\mathscr{R}$, and a scientific result $\theta(m)$ computed using measure m. The tests define a surviving set $\mathscr{M}^*$; the image $\Theta^* = \{\theta(m) : m \in \mathscr{M}^*\}$ reports the conclusions robust to measurement choice within the declared menu and validity argument. Held-out validity tests make the selection rule itself falsifiable. The framework is derived from first principles as partial identification over a finite menu of measurement functions, with maintained assumptions, a consistency proposition

for the plug-in admissible set and range, and a mandatory additive reporting layer for sampling uncertainty induced by data-dependent screening. It is implemented in the open, model-free package `cvprofiles` (2.5.0). Controlled synthetic data-generating processes illustrate the decision rule under known truth and verify the shipped holdout and coverage layers. Empirical results on real survey and AI measures are deferred until the preregistered study of Section 8 is executed on frozen inputs; a trust-and-preferences design is specified as that proof of concept. The broader research agenda is to make measurement selection auditable when candidate measures are abundant.

Keywords: measurement selection; construct validity; partial identification; AI measurement; synthetic cultural agents; cultural preferences; trust

1. When measures become cheap

Empirical science often begins with a concept that is scientifically meaningful but not directly observed. Trust, patience, reciprocity, ideology, polarization, and culture are familiar examples. Let C denote such a construct. Researchers learn about C through an operationalization m: a survey question, scale, experimental task, administrative proxy, factor score, text classifier, or other procedure that produces an observable score or response. The measure is not the construct. Its scientific value depends on whether the interpretation attached to its output is warranted.

For much of social science, producing a new measure was costly. Artificial intelligence relaxes this constraint. Large language models can label text, answer surveys, simulate respondents, generate scales, produce embeddings, and be adapted to represent target populations at low marginal cost [23, 2, 1, 28, 19]. The change in cost does not remove the measurement problem; it changes its location. If a prompt, embedding, composite, or fine-tuned model can be generated quickly, the scarce resource is no longer a candidate operationalization—it is evidence that the operationalization measures what the researcher means.

Recent audits of LLM use in social science find that model-generated measurements increasingly enter substantive analyses while validation practices remain heterogeneous [15]. At the 2026 Political Methodology meetings, generative AI was central in roughly a quarter of presentations, and most of that work was measurement-related [21]. Bauman et al. replicate 37 annotation tasks from 21 published studies and find that, with just a handful of prompt paraphrases, "virtually anything can be presented as statistically significant," with roughly 31% of published conclusions incorrect [3]. Model errors are correlated far beyond chance—same-wrong-answer agreement of 42% versus 13%, with better models *more* correlated—so agreement across models is weak evidence of validity [26], and LLM-based occupational-exposure scores are unstable across model choices [42]. Computational social science and NLP have begun importing construct-validity language explicitly [36, 22, 4]. Our premise is that this trend should be generalized beyond any one AI method: when measures are cheap, validity becomes the scarce resource.

That premise has a precise econometric home. Dell and Rambachan's 2026 NBER Summer Institute Methods Lectures—"Estimation and Inference with AI-Generated Data"—translate the Cronbach–Meehl nomological network into partial-identification language over a menu of candidate measures: moment-inequality restrictions, an admissible set, a construct-identified range for a downstream estimand, and an explicit treatment of estimation and inference (prediction-powered inference with validation samples, identification from repeated measurements, and data fusion across sources) [14, 12]. Their reading notes that additional restrictions can narrow the set of plausible measures, and the lectures close by calling for "principled, widely accepted methods" for construct validity. This paper operationalizes that call end-to-end: the decision rule, holdout-falsifiable selection, inference over the estimated admissible set, the open implementation, and a planned cultural application. The contribution is operational and auditable, not a claim to have discovered the bridge.

The closest 2026 neighbors share the destination but not the selection device. Chen, Rambachan, and Tamer derive an identified set for a downstream parameter under benchmark-informed restrictions on the correlation of measurement errors [11]; their restrictions bound error behavior, whereas ours are nomological-network moment inequalities over the menu, and the object of inference is the surviving set $\mathscr{M}^*$ rather than a single parameter. Oğut and Yin bound a regression coefficient using multiple nonlinear measurements, reaching the same $[\theta_L, \theta_U]$ destination without an admissible set or a selection rule [33]. Messing quantifies design-choice uncertainty in LLM evaluation pipelines, where naive standard errors are 40–60% too small—but the object is an evaluation score, not a construct measure, and admission uncertainty is absent [31]. Canen and Enamorado use multiple LLM measures as split-sample instruments to correct measurement error [9]; that is correction, not selection. None of these performs menu-level validation with a surviving set and a falsifiable selection rule.

A worked empirical workflow. Consider a researcher studying generalized interpersonal trust as a latent construct C, with scientific interest in an association between trust and a behavioral or economic outcome Y. A standard analysis begins from one operationalization—a survey item, a multi-item composite, a text score, or a model-generated proxy—and reports a point

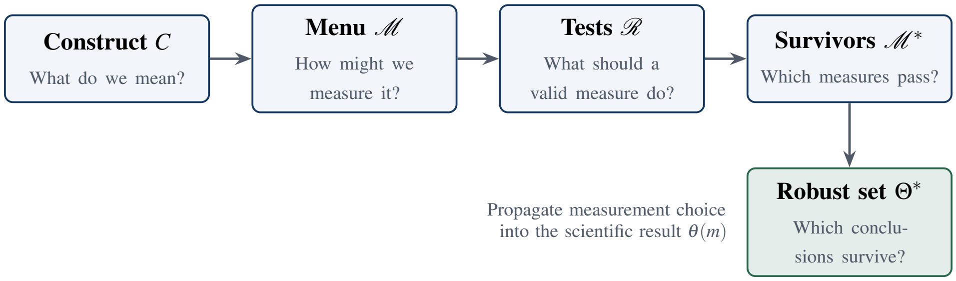

Figure 1: Core objects of the framework. The same five terms appear in prose, figures, and formal statements throughout.

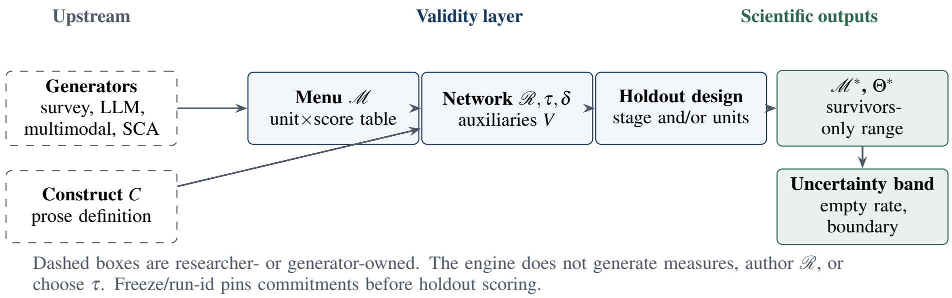

Figure 2: Epistemic systems view. Generators expand the menu; the validity layer adjudicates it; scientific conclusions are the image of a target functional on survivors, with additive sampling diagnostics.

estimate of $\theta(m) = \mathrm{Corr}(m, Y)$ or a related regression coefficient. When candidate operationalizations are cheap, the same design admits a *menu* $\mathscr{M}$ of such scores. Validity is not read off $\theta(m)$. It is read off auxiliaries V and a researcher-authored nomological network $\mathscr{R}(C)$: convergent association with related criteria, discriminant bounds against neighboring constructs, known-groups orderings, and other theory-implied inequalities, each with a substantive threshold τ_r chosen and justified by the researcher. Measures that satisfy the network form the surviving set $\mathscr{M}^*$; the image $\Theta^* = \{\theta(m) : m \in \mathscr{M}^*\}$ is the set of conclusions licensed under that validity argument. Some restrictions (or units) are held out so that the selection rule itself can fail. Threshold grids and related diagnostics stress-test τ without auto-tuning emptiness away. If a causal strategy with assumptions $\mathscr{A}$ yields an identified set $\Theta(m; \mathscr{A})$ for each fixed m, measurement-robust causal conclusions take the form $\bigcup_{m \in \mathscr{M}^*} \Theta(m; \mathscr{A})$—a composition of construct validation with identification, not a substitute for either.

2. One construct, many measures

The paper uses a deliberately small vocabulary. A *construct* C is the scientific concept of interest, defined in prose by its intended interpretation—not observed as a column. A *measurement function* is a map $m : \mathscr{X} \to \mathbb{R}$ from available inputs $X \in \mathscr{X}$ (survey responses, text, images, model states, ...) to a unit-level score, or a map to a vector followed by a *declared* scalar reduction that is part of the measure's definition. A *menu* $\mathscr{M} = \{m_1, \ldots, m_J\}$ is the finite set of such candidates declared by the researcher. *Auxiliaries* V are observed variables used to evaluate validity implications; they are not the latent construct. A set of *validity tests* (restrictions) $\mathscr{R}(C)$ encodes those implications as moment inequalities with researcher-chosen thresholds. Finally, $\theta(m)$ denotes the scientific result obtained when measure m is used in the downstream analysis.

The core workflow is

$$C \longrightarrow \mathscr{M} \xrightarrow{\mathscr{R}} \mathscr{M}^* \longrightarrow \Theta^*. \tag{1}$$

In words: define the construct, propose measures, test them, retain the surviving set, and ask which scientific conclusions survive measurement choice. Compatibility with the method is therefore *menu- and generator-agnostic over finite scalar (or scalarized) operationalizations*, not a claim that every conceivable measurement object is in scope without reduction.

Trust makes the distinction concrete. A researcher might consider

$$\mathscr{M}_{\text{trust}} = \{m_{\text{item}}, m_{\text{composite}}, m_{\text{base}}, m_{\text{persona}}, m_{\text{SCA2}}, m_{\text{noise}}\}. \tag{2}$$

The first two are conventional survey technologies; the next three are AI technologies; the last is a negative control. None is privileged by engineering complexity, human origin, or agreement with a preferred benchmark.

This perspective has a useful precedent in the Global Preferences Survey (GPS). The Preference Survey Module was designed by selecting and weighting survey items using their ability to predict incentivized experimental measures, then validating the resulting module across samples and countries [17, 18]. GPS is therefore not merely a source of target scores for synthetic agents. It is itself an instance of the underlying scientific problem: a construct is operationalized through a menu of candidate elicitation methods, and empirical evidence is used to choose among them. Our framework generalizes the menu beyond survey items and permits multiple validity criteria rather than a single criterion measure.

3. What should a valid measure do?

Construct validity is not a new problem. Cronbach and Meehl described validation through a nomological network [12]; Campbell and Fiske emphasized convergent and discriminant evidence across traits and methods [8]; later work emphasized that validity concerns the interpretation and use of scores [30, 25, 5].

For applied work, we translate these ideas into observable tests. Let V collect auxiliary variables and human evidence. A validity test $r \in \mathscr{R}(C)$ has a prespecified direction or threshold. A convenient general representation is the population moment inequality

$$r: \qquad \mathbb{E}\big[g_r\big(m(X), V; \tau_r\big)\big] \geq 0, \tag{3}$$

where τ_r is the *substantive threshold*: an exogenous, researcher-owned commitment—justified by theory, prior literature, or a preregistered tolerance—and *not* a parameter estimated by the admissibility procedure. Weakly positive inequalities are the normal form of one-sided validity content ("at least this association," "no more than this contamination," "at least this group gap"); equalities would overclaim exact magnitudes and empty $\mathscr{M}^*$ under sampling noise. Numeric gloss: $\tau_{\text{conv}} = 0.39$ means that, under the stated profile, any measure whose population correlation with the convergent criterion falls below 0.39 is denied the interpretation attached to C, regardless of how strongly it predicts Y. Examples include convergent association with a related measure, weak association with a theoretically unrelated construct, a known-groups difference, prediction of a behavioral criterion, or invariance across a declared nuisance dimension.

Validation as defined here is distinct from two adjacent activities [14]: *sensitivity analysis* asks whether conclusions change across procedures—the role of threshold grids over τ_r, of θ- and δ-grids, and of the uncertainty band in Section 5—whereas *hypothesis testing* characterizes sampling variability of a target $\theta(m)$ given a fixed measure. Validation links measures to an explicit criterion, the restrictions in $\mathscr{R}$; the layers are complementary, and the reporting protocol keeps their outputs separate.

Let $s_r(m)$ be the sample slack in test r (positive values indicate satisfaction). For tolerance $\delta \geq 0$, define the surviving set

$$\mathscr{M}^*(C; \mathscr{M}, \mathscr{R}, \delta) = \big\{m \in \mathscr{M} : s_r(m) \geq -\delta \;\; \forall r \in \mathscr{R}(C)\big\}. \tag{4}$$

We use *surviving set* (or *admissible set*) rather than "identified set" as the default language in the main text. Equation (4) is conditional on a researcher-supplied menu and a researcher-authored validity argument. A measure absent from $\mathscr{M}$ cannot be selected; a bad restriction in $\mathscr{R}$ can reject a useful measure or admit a poor one. The framework disciplines these commitments by making them explicit and auditable. It does not make them disappear.

Recent work such as the Construct Validity Protocol for embedding-based social measures provides concrete discriminant, incremental, and predictive tests [36]; Codebook LLMs provides a staged validation framework for LLM coding systems [22]; Licht et al. compare elicitation protocols for scalar constructs [27]. Our contribution differs in scope. The primitive is not a particular representation or model. It is the *menu*: survey items, composites, experiments, embeddings, prompts, and trained policies can enter together and face a common construct-level argument. The object of inference is $\mathscr{M}^*$, not any single m—menu-level validity rather than measure-level validity (Appendix A).

3.1 What passing means

If $m \in \mathscr{M}^*$, the warranted statement is conditional: m is consistent with the declared interpretation of C under the tests in $\mathscr{R}$ and the evidence used to evaluate them. Passing does not establish that $m = C$, that m causally measures C, that the menu is exhaustive, or that the measure is valid for uses not represented by the tests. An empty set is informative: it says the current menu and validity argument do not support the intended interpretation.

3.2 Maintained assumptions and consistency

Every application rests on four maintained assumptions, stated once and used everywhere (full statement in Appendix A):

(A1) *Menu coverage*—$\mathscr{M}$ contains the operationalizations of scientific interest (argued, never proven).

(A2) *Restriction validity*—each g_r is a correct testable implication of the construct definition and auxiliaries V (researcher-owned, falsifiable by data).

(A3) *Sampling*—the unit-by-measure score matrix is drawn under a stated sampling model.

(A4) *Tolerance policy*—$\delta \geq 0$ is declared ex ante (default 0) and is never tuned to rescue an empty set.

Any implication that could be wrong is treated as testable and belongs in $\mathscr{R}$, not among the maintained assumptions.

Under uniform consistency of slacks and of the target functional, and a separation condition that no measure lies exactly on the admissibility boundary, the plug-in admissible set and the plug-in range converge in probability to their population counterparts (Proposition A.1 in Appendix A). Boundary measures—those whose slack margins lie within estimation error of $-\delta$—are the documented failure regime: they flip in and out of $\mathscr{M}^*$ across samples and couple admissibility with the endpoints of the range. The reporting protocol monitors them rather than assuming them away.

4. Can validation itself be tested?

A validity framework can become circular if researchers inspect all available evidence, choose restrictions that favor a preferred measure, and then report that the measure passes. We therefore propose a held-out test of the selection rule itself. Two complementary splits implement that discipline; they answer different questions and must not be conflated.

Restriction-stage split. Partition the prespecified validity tests into

$$\mathscr{R} = \mathscr{R}_S \cup \mathscr{R}_H, \qquad \mathscr{R}_S \cap \mathscr{R}_H = \emptyset, \tag{5}$$

where S denotes selection and H denotes held-out evidence (same observational units). Construct the surviving set using only $\mathscr{R}_S$:

$$\mathscr{M}_S^* = \left\{ m \in \mathscr{M} : m \text{ passes every } r \in \mathscr{R}_S \right\}. \tag{6}$$

Then score $\mathscr{R}_H$ without letting those restrictions vote on admission. Holdout-stage failures are scientific *findings* about the selection rule; under a pure stage split they do not, by themselves, redefine the headline survivor set.

Units split (empirical core). Partition the sample into train and hold units (e.g., countries). Admit on the train frame; demand compliance of candidates on the hold frame. The robust set is

$$\mathscr{M}^*_{\text{robust}} = \mathscr{M}^*_{\text{select}} \cap \left\{ m : m \text{ passes the hold-frame screen} \right\}, \tag{7}$$

and the headline range is the image of θ on $\mathscr{M}^*_{\text{robust}}$ (with β evaluated on the full pooled sample over that set). Empty robust sets are valid findings. For the planned cultural application, country-level units-split is the primary falsification design; restriction-stage tags are the implementation vehicle and a secondary discipline against Goodharting the full network.

Falsifiable claim. A simple loss summarizes held-out violations,

$$L_H(m) = \sum_{r \in \mathscr{R}_H} w_r \, \ell\left\{ s_r(m) \right\}, \tag{8}$$

where ℓ penalizes negative or substantively small slack and w_r are declared weights. The falsifiable prediction is not that every survivor passes every unseen test. It is that construct-guided selection should improve unseen validity relative to relevant alternatives:

$$\mathbb{E}\left[L_H(m) \mid m \in \mathscr{M}_S^* \right] \; < \; \mathbb{E}\left[L_H(m) \mid m \in \mathscr{M}_B^* \right], \tag{9}$$

where $\mathscr{M}_B^*$ is a baseline set selected by a competing rule—best fit to a single anchor, best predictive fit on the selection sample, an LLM judge, or random/convenience selection. Baseline selection rules and pass-rate comparisons live in the evaluation harness; the Python API exposes stage tags, units-split composition, slacks, and holdout verdicts.

Equation (9) can fail. If it does, the proposed selection procedure has not shown that its validity language adds predictive scientific content. This held-out design is therefore central rather than decorative: it converts the framework from an organizing vocabulary into an empirical claim. In the open implementation (`cvprofiles` 2.5.0), the freeze/run-id contract hash-pins the network, anchors, stage split, holdout unit list, and seed before holdout scoring [13]—the operational meaning of "preregistered" here.

5. Which scientific conclusions survive?

Researchers usually care about an effect, association, ranking, forecast, or other scientific conclusion rather than the measurement score itself. Let $\theta(m)$ denote that result when measure m is used. Once the surviving set is defined, report

$$\Theta^*(C; \mathscr{M}, \mathscr{R}) = \left\{ \theta(m) : m \in \mathscr{M}^* \right\}. \tag{10}$$

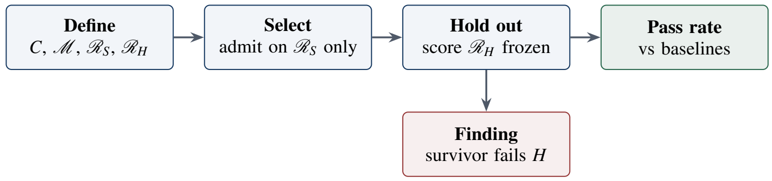

Figure 3: Holdout workflow that makes the selection rule itself falsifiable (Section 4).

For a scalar target, a transparent summary is the construct-identified range

$$[\theta_L, \theta_U] = \Big[\min_{m \in \mathscr{M}^*} \theta(m),\ \max_{m \in \mathscr{M}^*} \theta(m)\Big]. \tag{11}$$

We call Θ^* the *robust result set*. It is not an unconditional identified set for the latent construct. It is the set of results produced by measures that survive the declared validity argument [29, 39, 32].

This step is related to, but distinct from, multiverse and specification-curve analysis [40]. Specification curves ask how an estimate varies across theoretically justified analytical specifications. Our ordering is different: first ask which operationalizations survive evidence about the *meaning* of the construct; then propagate measurement choice among those survivors. The approaches can be combined—for example, a specification curve within each $m \in \mathscr{M}^*$, reporting uncertainty over both measurement and analysis choices.

The distinction matters empirically. If a single survey item, a PCA score, and an AI policy all purport to measure trust, choosing one before estimating a trust–development relationship hides a source of uncertainty that conventional standard errors do not capture. Equation (10) makes that uncertainty visible. A narrow Θ^* supports a result robust to surviving measurement choice; a wide or sign-changing Θ^* is a substantive finding about the fragility of the scientific conclusion.

5.1 Inference over the admissible set

The headline object remains the plug-in range (11): the image of θ on $\mathscr{M}^*$, with no model and no shrinkage. Because $\mathscr{M}^*$ is estimated, admission and endpoints are coupled whenever measures lie near the boundary. The following reporting layer is *mandatory and additive* (it does not replace the headline range); `cvprofiles` 2.5.0 implements items (1)–(3) under synthetic gates:

1. **Uncertainty band.** Units-resampling bootstrap recomputes slacks, $\hat{\mathscr{M}}^*$, and $[\hat{\theta}_L, \hat{\theta}_U]$ per replicate (seeded). Report per-side quantiles at $\alpha/2$ ($\alpha = 0.10$ default) over non-empty replicates. Label the object an *uncertainty band*—selection uncertainty on the pooled sample, not a formal confidence region under arbitrary dependence, and not a holdout-robustness band [24, 41].
2. **Empty-replicate rate.** The share of replicates with empty $\hat{\mathscr{M}}^*$ is reported explicitly; the band is conditional on non-empty admissible sets.
3. **Boundary attribution.** Flag measures whose absolute slack margin satisfies $|\text{margin}_m| \le \kappa \cdot \text{SE}_m$ ($\kappa = 2$ default): near-threshold admissions and near-miss rejections are fragile; far-rejected measures are not.
4. **Conservative cross-check (optional).** Per-admissible-measure intervals with a Bonferroni adjustment over $|\hat{\mathscr{M}}^*|$, unioned around the plug-in range; if the projection is materially wider than the band, the coupling is material.

Full simultaneous-coverage theory for the coupled band is deferred to the companion methods paper; the present paper commits to the reporting protocol above (Appendix A).

6. AI as a measurement generator

The framework deliberately separates generating a measure from validating it. Four layers of empirical work should not be collapsed into one:

(i) *Hypothesis generation*—proposing constructs, menus, and theory fragments (including agent-assisted exploration);

(ii) *Measurement*—maps $m : \mathscr{X} \to \mathbb{R}$ (or declared scalarizations) that fill the unit-by-measure table;

(iii) *Construct validation / menu-level partial identification*—restrictions $\mathscr{R}$, survivors $\mathscr{M}^*$, and robust conclusions Θ^*;

(iv) *Estimation and inference*—sampling uncertainty about data-dependent screening, and downstream causal or predictive analysis composed with $\mathscr{M}^*$.

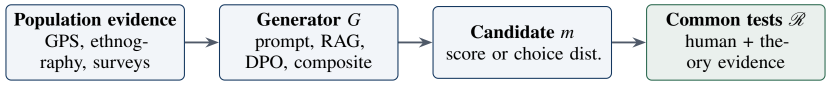

Figure 4: AI procedures and conventional instruments enter as generators of candidate measures; the common validity layer determines the interpretation each candidate can support.

Let

$$G(D,C) \longrightarrow m \tag{12}$$

denote a measurement-generation technology that uses evidence D and a target construct C to produce a candidate measure m. This notation places conventional and AI procedures—including multimodal models that map high-dimensional observations to scores—on equal epistemic footing without claiming they have the same engineering or causal properties. Generators expand the menu; they do not adjudicate it.

A base language model provides a candidate through unconditioned elicitation; persona prompting supplies a target identity at inference; ethnographic retrieval enriches that context [10]. These are *contextual projections*: model weights are fixed while the information supplied at inference changes. Preference fine-tuning moves population signal into an estimated policy [37, 35, 34]. Inference-time steering offers a third route [16]. In all cases, the object that enters empirical analysis is the evaluated output $m(X)$, so each belongs in the same measurement menu.

We avoid the claim that moving information into weights makes the resulting model "structural" in the econometric sense. The defensible claim is narrower: trained policies produce persistent, estimated preference representations whose external validity can be compared with prompting and conventional measures under a common validity argument. Under a Bradley–Terry reading of preference optimization, option probabilities form a discrete-choice distribution with reward margins identified relative to a reference policy [6, 37]; what is identified from *aggregate* population moments—as opposed to item-level responses—is a set of choice distributions consistent with those moments, not cardinal utility or within-unit heterogeneity (Appendix B).

Synthetic Cultural Agents are therefore one family of measurement-generation technologies for culture [10]. Culture is the demanding application through which the methods are developed; measurement selection is the general methodological contribution. The same generator–validation distinction applies to sentiment, ideology, norms, political preferences, and other latent constructs.

7. How the engine works: controlled synthetic illustrations

Before any real-world claim, the decision rule should be transparent on data-generating processes (DGPs) whose truth is known by construction. This section uses *population-limit* examples—exact correlations and group gaps implied by a linear factor structure—so that every slack and every admission decision can be verified by hand. No Monte Carlo noise enters the headline numbers in Tables 1–2. The examples use restriction types and target functionals available in `cvprofiles` 2.5.0 (`corr_min`, `corr_zero`, `mean_order`, and `corr_y`, among others) [13]. A finite-sample engine check under the same design is reported briefly after DGP D; it verifies shipped holdout and coverage behavior and does *not* replace the closed forms.

7.1 Setup

Units are exchangeable draws from a latent construct C with $\mathrm{Var}(C) = 1$. Auxiliaries are a convergent criterion V_c, a discriminant foil V_d, a binary known-groups indicator G, and an outcome Y used by the target functional $\theta(m) = \mathrm{Corr}(m,Y)$. The menu is

$$\mathscr{M} = \{m_{\text{true}}, m_{\text{weak}}, m_{\text{contam}}, m_{\text{method}}, m_{\text{noise}}\}, \tag{13}$$

with population loadings given in Table 1. All variables are standardized. Under a linear one-factor structure, population correlations equal products of loadings (plus contamination terms where stated).

Population design for auxiliaries (also standardized): $\mathrm{Corr}(V_c,C) = 0.80$, $\mathrm{Corr}(V_d,C) = 0$, $\mathrm{Corr}(Y,C) = 0.60$, and $\mathbb{E}[C \mid G{=}1] - \mathbb{E}[C \mid G{=}0] = 0.80$ (known-groups gap on the latent). Measure–group gaps scale with the loading on C.

7.2 DGP A — clean separation

Selection network $\mathscr{R}_S$ (three restrictions, $\delta = 0$):

$$r_{\text{conv}}: \ \mathrm{Corr}(m,V_c) \geq 0.39, \tag{14}$$

$$r_{\text{disc}}: \ \left|\mathrm{Corr}(m,V_d)\right| \leq 0.25, \tag{15}$$

$$r_{\text{grp}}: \ \mathbb{E}[m \mid G{=}1] - \mathbb{E}[m \mid G{=}0] \geq 0.25. \tag{16}$$

Table 1: Population loadings for the synthetic menu. C is the latent construct; V_d is a discriminant foil orthogonal to C in the population. "Method" denotes idiosyncratic method variance.

Measure	Loading on C	Loading on V_d	Method variance	Intended role
m_{true}	0.90	0	residual	High-quality operationalization
m_{weak}	0.50	0	residual	Valid but noisy
m_{contam}	0.70	0.55	residual	Contaminated by discriminant foil
m_{method}	0.20	0	dominant	Method-variance dominated
m_{noise}	0	0	1	Negative control

Table 2: DGP A (clean separation). Population slacks for $\mathscr{R}_S$ and target $\theta(m) = \text{Corr}(m,Y)$. Admission requires all three slacks ≥ 0. Numbers are exact population quantities under Table 1, not simulation estimates.

Measure	s_{conv}	s_{disc}	s_{grp}	$\theta(m)$	In $\mathscr{M}^*$?
m_{true}	$+0.33$	$+0.25$	$+0.47$	0.54	yes
m_{weak}	$+0.01$	$+0.25$	$+0.15$	0.30	yes
m_{contam}	$+0.17$	-0.30	$+0.31$	0.42	no (r_{disc})
m_{method}	-0.23	$+0.25$	-0.09	0.12	no (r_{conv}, r_{grp})
m_{noise}	-0.39	$+0.25$	-0.25	0.00	no (all but disc.)

Target: $\theta(m) = \text{Corr}(m,Y)$.

Exact population slacks and admission decisions appear in Table 2. Only m_{true} and m_{weak} survive. The robust result set is

$$\Theta^* = \{\text{Corr}(m_{\text{true}},Y), \text{Corr}(m_{\text{weak}},Y)\} = \{0.54, 0.30\}, \qquad [\theta_L, \theta_U] = [0.30, 0.54]. \tag{17}$$

Note that every admissible slack is strictly positive under $\tau_{\text{conv}} = 0.39$ (separation holds). DGP B isolates the knife-edge case $\tau_{\text{conv}} = 0.40$, where m_{weak} sits exactly on the convergent boundary, and shows why boundary measures couple admission with the range. The contaminated measure fails discriminant validity; the method-heavy and pure-noise measures fail convergent and known-groups tests. An empty set does not arise: the menu and network are compatible.

7.3 DGP B — boundary measures and coupling

Raise the convergent threshold to $\tau_{\text{conv}} = 0.70$, holding the rest fixed. Population correlations with V_c are still 0.72, 0.40, 0.56, 0.16, and 0 (loadings times 0.80). Then m_{true} remains admissible ($s_{\text{conv}} = 0.72 - 0.70 = +0.02$), while m_{weak} falls to $s_{\text{conv}} = 0.40 - 0.70 = -0.30$ and is rejected. The plug-in range collapses to the singleton $\{0.54\}$.

If, instead, the threshold is set at the weak measure's population correlation with V_c ($\tau_{\text{conv}} = 0.40$), then $s_{\text{conv}}(m_{\text{weak}}) = 0$ exactly: m_{weak} lies *on* the admissibility boundary—the knife-edge regime the boundary-attribution protocol of Section 5 is designed to monitor. In finite samples the plug-in decision for m_{weak} is decided by sampling noise, and because $\theta(m_{\text{weak}})$ is the lower endpoint of the range whenever it is admitted, admission error and endpoint error are coupled. `cvprofiles` 2.5.0 recomputes $\mathscr{M}^*$ inside a units bootstrap, reports an empty-replicate rate, labels an additive uncertainty band, attributes boundary measures via $|\text{margin}_m| \leq \kappa \cdot \text{SE}_m$, and its Python API exposes both restriction-stage and units-split holdout. Those layers are required before real-world results enter this paper; they are no longer missing engine primitives.

7.4 DGP C — empty $\mathscr{M}^*$ is a finding

Replace r_{disc} with a conflicting requirement $\text{Corr}(m,V_d) \geq 0.50$ and raise the convergent threshold to $\tau_{\text{conv}} = 0.60$, keeping r_{grp} at 0.25. No measure in Table 1 can load heavily on both C (hence on V_c at ≥ 0.60) and the orthogonal foil V_d (at ≥ 0.50) at once: m_{true} passes convergent ($0.72 - 0.60 = +0.12$) but fails the foil requirement ($0 - 0.50 = -0.50$); m_{contam} passes the foil ($0.55 - 0.50 = +0.05$) but narrowly fails convergent ($0.56 - 0.60 = -0.04$). The admissible set is empty. Under the framework this is not a software failure: it is a rejection of the pair $(\mathscr{M}, \mathscr{R})$—the proposed operationalizations together with the restrictions asserted for them—not a verdict on the construct C itself [13]. The honest response is to revise the menu or the network, never the tolerance δ.

7.5 DGP D — holdout discipline (non-trivial closed form)

To illustrate Eq. (9) under known truth, use a lenient selection screen and a stricter holdout screen on the same loadings. Let

$$\mathscr{R}_S = \{\text{Corr}(m,V_c) \geq 0.40,\ |\text{Corr}(m,V_d)| \leq 0.60\}, \tag{18}$$

$$\mathscr{R}_H = \{|\text{Corr}(m,V_d)| \leq 0.20,\ \mathbb{E}[m \mid G{=}1] - \mathbb{E}[m \mid G{=}0] \geq 0.25\}. \tag{19}$$

	m_{true}	m_{weak}	m_{contam}	m_{method}	m_{noise}
r_{conv}	+0.33	+0.01	+0.17	−0.23	−0.39
r_{disc}	+0.25	+0.25	−0.30	+0.25	+0.25
r_{grp}	+0.47	+0.15	+0.31	−0.09	−0.25
admit?	yes	yes	no	no	no

Green: slack ≥ 0 (pass). Red: slack < 0 (fail). Bottom row: membership in $\mathscr{M}^*$.

Figure 5: Visual slack profile for DGP A. The surviving set is the set of columns with an all-green restriction profile.

Under Table 1, selection admits three measures: m_{true} (slacks $+0.32$, $+0.60$), m_{weak} ($+0.00$, $+0.60$), and m_{contam} ($+0.16$, $+0.05$)—the lenient discriminant bound lets contamination through. On holdout, m_{contam} fails the tight discriminant test ($s = 0.20 - 0.55 = -0.35$), while m_{true} and m_{weak} pass both holdout restrictions. The validity-guided holdout pass rate is therefore $2/3 \approx 0.67$.

An anchor-fit baseline that keeps the two measures with largest $|\text{Corr}(m, V_c)|$ retains $\{m_{\text{true}}, m_{\text{contam}}\}$. Only one of those two survives $\mathscr{R}_H$, so the baseline pass rate is $1/2 = 0.50$. In this population-limit example, construct-guided selection beats single-anchor selection on held-out validity—exactly the direction of Eq. (9). The lenient-then-strict discriminant thresholds are intentional: they model ex-ante uncertainty about how tightly a discriminant bound should be set, which is precisely the kind of threshold judgment that a holdout partition is designed to stress-test rather than to hide inside a single-stage screen. The operational claim of the paper is that the same comparison, on real data with a preregistered split, is the empirical core once frozen inputs and the evaluation harness execute Eq. (9) end-to-end. `cvprofiles` 2.5.0's Python API already exposes stage-tagged restrictions, units-split composition, and holdout verdicts under synthetic gates.

7.6 Finite-sample engine check (not headline numbers)

To verify that the shipped pipeline reproduces the qualitative structure of DGP A–D, we drew n=2000 standardized units from the linear factor design of Table 1 (seed 20260808) and ran `cvprofiles` 2.5.0 with bootstrap (n_{boot}=400, α=0.10, κ=2). Under the DGP A network, the admissible set was $\{m_{\text{true}}, m_{\text{weak}}\}$ with plug-in range approximately [0.285, 0.522] (population [0.30, 0.54]). The uncertainty band was approximately [0.249, 0.550] with empty-replicate rate 0; only m_{weak} was flagged as boundary, with admission frequency $\hat{p}_{\text{weak}} \approx 0.95$ and $\hat{p}_{\text{true}} = 1$. Under the DGP D stage split, selection admitted $\{m_{\text{true}}, m_{\text{weak}}, m_{\text{contam}}\}$ and the holdout verdict failed m_{contam} on the tight discriminant restriction—matching the closed-form direction. A units-split on the DGP A network retained the same robust pair. These runs confirm engine behavior; they do not replace the hand-verified population tables above. (Local engine check under the P5 implementation, seed 20260808: DGP A run_id `271b6d98...`, stage-split `3a6c8f8a...`, units-split `8508062e...`; verification script `tools/paper_synth_p4p5.py` in the `cvprofiles` repository. Qualitative structure matches the closed forms; run-ids are diagnostic, not population quantities.)

7.7 What the illustrations show—and what they do not

The synthetic DGPs show that the decision rule is well-defined, that empty and wide sets are expressible findings, that boundary measures are the fragile case, and that holdout splits change the scientific question from "does my favorite measure pass?" to "does my selection rule predict unseen validity?" They do *not* show that real survey items, persona prompts, or preference-trained policies survive any particular network for trust or culture. That is the content of Section 8, not of this section.

8. Applied illustrations: from observation to falsification

Empirical cells in this manuscript are empty by design. The coupled-inference reporting layer, holdout-restriction workflow, and evaluator coverage required by Sections 4–5 now ship in `cvprofiles` 2.5.0 under synthetic gates [13]. What remains before empirical cells are licensed is (i) frozen, preregistered inputs, (ii) an evaluation harness that implements the baselines of Eq. (9), and (iii) execution of the studies described here on real survey and AI measures. The flagship illustration takes the cultural-preference application end to end—from observation of population moments to the falsification of a measure-selection rule—while a second, deliberately shorter illustration applies the same machinery to AI benchmarks. This section preregisters both.

Table 3: Cultural-preference menu for the flagship illustration. The AI arms—persona prompting and the SCA 2 adapter—are the scientific focus; conventional instruments and the noise control anchor the common screen.

Candidate	Information source	Scientific role
Single survey item	Direct human response	Conventional low-cost operationalization; interpretable but potentially narrow.
Composite / PCA / IRT	Multiple human responses	Aggregates information; weights may alter construct meaning.
Base prompting	Pretrained model	Tests structure already present without cultural conditioning.
Cultural / persona prompt	Pretrained model + declared culture	Direct identity at inference: the population is named in the prompt; standard inference-time synthetic-subject baseline.
SCA 2 adapter	Pretrained model + GPS-anchored preference training	Indirect encoding: the country is never named; preference representation is trained, not prompted.
Noise / shuffled control	No valid construct signal	Negative control; a useful framework should reject it.

8.1 Flagship illustration: cultural preferences, SCA 2 versus persona prompting

The flagship application compares two AI measurement technologies—persona prompting and the SCA 2 adapter—against conventional instruments for cultural preferences, under one common validity argument, from observation to falsification.

Observation. The observation surfaces are population moments: country-level and individual-level GPS profiles, standardized as deviations from the global mean [17], and Wave 7 WVS and AmericasBarometer moments for the USA and Mexico, with items pre-registered into evaluation (tier 2) and held-out validation (tier 3) sets.

Construct C: cultural preferences across the six GPS dimensions; generalized interpersonal trust—the disposition to expect cooperative behavior from anonymous others in the absence of contractual enforcement—serves as the primary construct, defined independently of any single survey instrument [20, 38, 7].

Menu (Table 3): the six-method design of Eq. (2), spanning conventional and AI generators plus a noise control. The two AI arms are the scientific focus: a cultural persona prompt, which declares the target population at inference, and the SCA 2 adapter, whose preference representation is trained from GPS-anchored labels while the country is never named—indirect encoding, so any population-specific behavior must be learned rather than prompted [10]. All candidates face the same declared validity tests; the table does not presume which will survive.

8.2 Validity profile and holdout split

The validity profile will include, at minimum: convergent evidence from related trust measures; discriminant evidence against neighboring but distinct constructs (e.g., institutional trust, risk preference); known-groups or demographic gradients declared from prior literature; external behavioral or economic criteria where available; cross-country ranking where justified; negative controls; and, where items are administered to both human and model respondents, human–model invariance tests (differential item functioning between humans and chatbots as a declared restriction) [47]. Wherever possible, tests are partitioned into $\mathscr{R}_S$ and $\mathscr{R}_H$ under the pre-data discipline of Section 4. Two holdout devices implement falsifiability: tier 3 WVS moments, never used in training or scoring, are reserved as holdout restrictions; and the units-split assigns the USA and Mexico to separate frames, admitting on the training frame and demanding compliance on the held frame, with the robust set as headline (Section 4). Thresholds τ_r are literature-anchored and frozen before holdout scoring.

8.3 Downstream target and robust result set

Primary $\theta(m)$: association of the trust measure with a preregistered economic or behavioral criterion (and, secondarily, cross-country ranking fidelity where the criterion is defined). Report $\mathscr{M}^*$, Θ^*, the uncertainty band, empty-replicate rate, and boundary set. Empty or wide sets are admissible findings.

8.4 Baselines for the falsifiable core

Compare validity-guided selection to at least: (i) best fit to a single anchor (e.g., maximize correlation with GPS trust); (ii) best predictive fit on selection-sample criteria; (iii) random or convenience selection among AI measures, which serves as the default floor: a selection rule that cannot beat random selection on held-out validity carries no predictive content. The headline scientific claim of the empirical paper is Eq. (9), not that any one generator "wins."

Table 4: Evidence matrix. Empirical cells are empty in this version; the matrix is the preregistration of what must be shown.

Claim	Required evidence	Closest comparison	What would weaken it
Heterogeneous measures share one construct-level screen	Common benchmark: survey, composite, persona prompt, SCA 2 adapter, noise	Psychometrics; Proxy Presumption; Codebook LLMs	Tests require incompatible method-specific definitions or fail to reject noise
Validity-based selection has predictive content	Preregistered $\mathscr{R}_S/\mathscr{R}_H$ across constructs/populations	Cross-validation; predictive model selection	Survivors do no better on holdout than anchor-fit, prediction, or random
Measurement choice changes conclusions	Θ^* for consequential analyses	Specification curves; multiverse	Surviving measures always yield indistinguishable results
AI and conventional measures face the same standard	Head-to-head cultural-preference benchmark (USA/MEX)	LLM measurement frameworks	Surfaces not meaningfully comparable across technologies
SCA 2 versus persona prompting is decided by evidence, not engineering	Preregistered units-split (USA/MEX) + tier 3 holdout moments; baselines: random, anchor-fit, LLM judge	Persona simulation; cultural fine-tuning; Codebook LLMs	Survivors do no better on held-out moments than persona prompting or random selection
Benchmark items as a menu	Construct-validated benchmarks: task-family menu, cross-benchmark foils, held-out tasks	Benchmark audits; construct-validity reviews	Scores do not separate from foils, or held-out task families fail to predict benchmark ranks

8.5 Secondary illustration: AI benchmarks as a measurement menu

A benchmark score is a measure: it maps model outputs to a number interpreted as ability, safety, or knowledge. The same machinery applies with benchmark families as the menu—reasoning suites, safety evals, cultural-knowledge benchmarks—and other benchmarks as convergent and discriminant foils in the validity network. Held-out task families play the role of $\mathscr{R}_H$, giving benchmark construct validation an anti-contamination design: an advertised capability should survive on tasks a model never saw [4]. The motivating fragility is documented: the Stanford HAI 2026 AI Index reports that nearly half of the sixty most-cited benchmarks are saturated at the frontier [48], contamination is best treated as a spectrum rather than a flag, model errors are correlated far beyond chance [26], and prompt paraphrases can make virtually anything statistically significant [3]. This illustration is deliberately less developed: it is included to show that the menu-level screen is not specific to cultural measurement, not to claim a benchmark result in this version.

8.6 Extension ladder (not required for the first lock)

1. Multi-country coverage spanning preference space and baseline-model proximity.
2. All six GPS dimensions as a generalization exercise: for each dimension k, ask whether $m_{\mathrm{SCA2},k} \in \mathscr{M}_k^*$ [17].
3. Composition with causal assumptions: if $\Theta(m;\mathscr{A})$ is a causal identified set under measure m and assumptions $\mathscr{A}$, report $\bigcup_{m\in\mathscr{M}^*}\Theta(m;\mathscr{A})$ as measurement-robust causal conclusions [28]—a composition, not an identification of construct validity with causality.
4. Human–model measurement invariance: where items are administered to both humans and models, test invariance (DIF) as a declared restriction rather than an assumption [47].

8.7 What would weaken each claim

Table 4 is the working evidence matrix for a general-science submission. The failure column states what would force a narrower claim.

9. Results

No empirical results are reported in this version. Preliminary two-country adapter comparisons and standalone survey-correlation exercises are deliberately omitted. They cannot support the intended claims until the preregistered trust study of Section 8 is executed on frozen inputs with an evaluation harness that implements Eq. (9). Engine primitives for holdout and the additive uncertainty band are no longer the binding constraint; real-world evidence is. Reporting partial real-world numbers now would invite over-interpretation of undisciplined evidence. Synthetic illustrations (Section 7) demonstrate the decision rule under known truth; they are not substitutes for the real-world proof of concept.

10. Implementation and release strategy

`cvprofiles` is the companion implementation of the measurement-selection layer [13]. Its design goal is intentionally modest: accept a frozen unit-by-measure score table, a researcher-authored set of validity tests, thresholds, and downstream functionals; return a full measure-by-test profile, the surviving set, the robust result set, and sensitivity diagnostics. It does not call an LLM, choose the construct, write the validity argument, or decide what scientific threshold is acceptable. This separation keeps the method auditable and makes the software usable outside AI research.

v2.5.0 (shipped under synthetic gates; not a PyPI release). Score → restrict → identify → report; survivors-only range; empty $\mathscr{M}^*$ as exit-success finding; freeze/run-id discipline; additive bootstrap, θ- and δ-grids; expanded registry (`corr_min`, `corr_sign`, `corr_zero`, `mean_order`, `rank_agree`, `monotone_rank`; betas including `corr_y`, `ols_coef`, `diff_means`, `map_distance`); restriction-stage and units-split holdout with robust headline; additive uncertainty band with empty-replicate rate and boundary attribution $|\text{margin}_m| \leq \kappa \cdot \text{SE}_m$; model-free engine.

Empirical license (still missing, by design). Locked trust construct and validity profile; six-method common menu on frozen inputs; held-out validity evidence with baseline comparisons for Eq. (9); negative controls; robustness across seeds and training choices where AI measures enter; frozen artifacts sufficient for third-party replication. Public model release should follow rather than substitute for the common benchmark. The strongest near-term use of synthetic cultural agents remains hypothesis generation and instrument design—identifying behavioral margins worth testing with humans—rather than replacing human evidence.

11. What the framework does—and does not—claim

1. **A surviving measure is not the construct.** Passing $\mathscr{R}$ supports a stated interpretation under stated evidence; it does not establish ontological identity or a unique causal measurement relation.
2. **The menu is not exhaustive.** $\mathscr{M}^*$ is conditional on $\mathscr{M}$. Adding a new survey, experiment, model, or estimator can change the survivor set and the robust result set.
3. **Validity tests are scientific assumptions with observable content.** They should be justified from theory and prior evidence, declared before inspecting benchmark outcomes when feasible, and stress-tested over plausible thresholds.
4. **Measurement validity is use-specific.** A policy that reproduces country rankings may still be unsuitable for individual prediction, absolute population shares, or causal counterfactuals.
5. **AI agreement is not independent validation.** Several LLM procedures may share training data, priors, and failure modes [8]. Human, behavioral, and different-method evidence remain essential.
6. **The feedback loop is falsifiable, not universally verifiable.** Validity tests encode substantive commitments and remain open to revision; an agent optimizing only to enter $\mathscr{M}^*$ can Goodhart the restrictions. We speak of auditable or theory-grounded feedback, not of a closed verifiable reward for autonomous science.

These boundaries are features rather than weaknesses. Empty survivor sets, wide robust result sets, and failed held-out tests are scientifically meaningful outcomes.

12. Discussion: measurement validation and the automation of social science

The four-layer separation of Section 6—hypothesis generation, measurement generation, construct validation, and estimation/inference—is more than bookkeeping; it is a claim about where scientific labor can and cannot be delegated. Generators are cheap and increasingly agentic: they propose constructs, draft menus, and fill the unit-by-measure table, while estimation is, in most applications, solved engineering. The layer that resists automation is the gate between them—construct validation, where a researcher-authored network $\mathscr{R}$, thresholds τ_r, and a held-out design adjudicate which measures may carry the interpretation attached to C. Validation is the only layer whose failure is itself a scientific output: an empty $\mathscr{M}^*$, a wide Θ^*, or a failed holdout verdict revises the research program, whereas an architecture that collapses the layers inherits the failure modes of its weakest component.

Proposals for automated social science make the stakes concrete. Manning, Zhu, and Horton suggest that language models can serve as both scientists and subjects, generating hypotheses and simulating respondents [28]. That program is attractive precisely because the first and fourth layers—generation and estimation—are what models do best; what such loops lack is the third layer, an independent gate that can fail. The difficulty is not new in kind. Reinforcement learning with verifiable rewards (RLVR-style training) discovered the same architecture the hard way: when the verifier is learned or adjustable, the generator optimizes the verifier rather than the property, and neural process-reward models have been shown to score invalid proofs highly—fluency detectors more than reasoning verifiers. The field's response has been to move verification toward checks that cannot be learned away: frozen, private, or human-committed evaluation sets, a turn sharpened by recent analyses of public versus private benchmarks [43]. The holdout-falsifiable validity commitment of Section 4 is the measurement-side analogue: restrictions $\mathscr{R}_H$, thresholds, and units are frozen before scoring, so the selection rule itself can fail. The honest caveat is that this verifiability is social-contractual rather than computational—human-frozen and community-adjudicated—so we claim auditable, theory-grounded feedback, never a closed verifiable reward (Section 11, point 6).

The 2026 critique literature arrives at the same conclusion from the measurement side. LLM-simulated experiments are observational studies under user-drift confounding: the researcher controls the prompt, and comparisons across conditions are identified only if user-driven variation is held fixed [44]. Heuristic substitution of simulated for human evidence—treating model outputs as interchangeable with responses—cannot support confirmatory claims [45], and shadow evaluations, in which frontier agents work an unpublished research question and the original authors grade the output, find that agentic pipelines fall short of expert research under scrutiny that verifier-scored tasks and blind review cannot provide [46]. Each critique names the same missing object: an ex ante, falsifiable validation record. The framework of this paper is that record. It does not prevent user drift or heuristic substitution; it converts them into testable restrictions—drift-sensitive auxiliaries, discriminant bounds, negative controls—whose failure is reported rather than absorbed.

What would make an automated loop trustworthy? The commitment order matters more than the automation: the construct definition, the network $\mathscr{R}$, the thresholds τ_r, and the holdout design must be frozen before any scoring, and the revision loop must be driven by the framework's own failure signals—empty $\mathscr{M}^*$, wide Θ^*, failed holdout verdicts—which are feedback to the researcher, not permissions for an agent. Automation is safe to the extent that it is confined to the layers the framework does not gate, while the gate itself remains human-owned. Fully autonomous workflows therefore remain a contingency rather than a contribution: until `cvprofiles` 2.5.0 is operational on real frozen inputs under the preregistered study of Section 8, the claims of this paper concern the validation gate, not its replacement.

13. Conclusion: measurement when generation is no longer scarce

Science does not become easier merely because candidate measures become cheap. Abundance moves the bottleneck from producing an operationalization to justifying one. The appropriate response is not to privilege human surveys over AI, or fine-tuning over prompting, but to make competing measures answer to the same construct-level evidence.

The proposed framework asks five questions in order: What do we mean? How might we measure it? What should a credible measure do? Which measures survive? Which scientific conclusions survive with them? Formally,

$$C \longrightarrow \mathscr{M} \xrightarrow{\mathscr{R}} \mathscr{M}^* \longrightarrow \Theta^*.$$

Dell and Rambachan supplied the partial-identification reading of the nomological network and the call for principled construct-validity methods [14]; this paper supplies the operational end-to-end layer, the open tool, the controlled illustrations, and the preregistered path to a real-world proof of concept.

For the EconLLM research program, Synthetic Cultural Agents are measurement-generation technologies for culture. Neither prompting nor preference training is validated by construction. Both become scientifically useful to the extent that their outputs survive independent evidence about the constructs they claim to represent. When measures are cheap, validity becomes the scarce resource.

A. First-principles formalization

This appendix re-derives the framework from first principles as a robustness check on the main-text definitions. Notation matches Sections 3–5.

A.1 Primitives

Definition A.1 (Measurement menu)**.** *A* measurement function *is a map $m : \mathscr{X} \to \mathbb{R}$ (or to a finite choice simplex) producing a score or response distribution from available inputs $X \in \mathscr{X}$. A* menu *is a finite collection $\mathscr{M} = \{m_1, \ldots, m_J\}$ declared by the researcher before inspecting the target estimand.*

Definition A.2 (Nomological restriction)**.** *Given auxiliaries V and a threshold $\tau_r \in \mathbb{R}$, a* restriction *is a functional inequality $\mathbb{E}[g_r(m(X), V; \tau_r)] \geq 0$. The* sample slack *$s_r(m)$ is a sample analogue of the left-hand side (or a monotone transform thereof); the sign convention is $s_r(m) \geq 0$ when the restriction is satisfied.*

Definition A.3 (Admissible set and robust result set)**.** *For tolerance $\delta \geq 0$, $\mathscr{M}^* = \{m \in \mathscr{M} : s_r(m) \geq -\delta \ \forall r \in \mathscr{R}\}$. For a target functional $\theta : \mathscr{M} \to \mathbb{R}$, $\Theta^* = \{\theta(m) : m \in \mathscr{M}^*\}$ and $[\theta_L, \theta_U] = [\min \Theta^*, \max \Theta^*]$ when $\mathscr{M}^* \neq \emptyset$.*

Definition A.4 (Menu-level validity)**.** *Inference is* menu-level *when the object of scientific interest is $\mathscr{M}^*$ (and functionals of $\mathscr{M}^*$), not a single preferred m. Per-measure protocols [36, 22] validate individual operationalizations; they are complementary inputs to a menu, not substitutes for menu-level inference.*

A.2 Maintained assumptions

Assumption A.1 (Menu coverage)**.** *$\mathscr{M}$ contains the operationalizations of interest for the research question. Coverage is argued from literature and design; it is never proven.*

Assumption A.2 (Restriction validity)**.** *Each g_r is a correct testable implication of the construct definition and auxiliaries V. Restrictions are researcher-owned and falsifiable by data.*

Assumption A.3 (Sampling)**.** *The unit-by-measure score matrix is drawn under a stated sampling model (e.g., exchangeable country-level aggregates), frozen with the run.*

Assumption A.4 (Tolerance policy)**.** *$\delta \geq 0$ is declared ex ante (default 0) and is never tuned to rescue an empty set.*

Maintained vs. testable. Assumptions above are stated once and used everywhere. Any implication that could be wrong is testable and belongs in $\mathscr{R}$.

A.3 Consistency of the plug-in objects

Proposition A.1 (Consistency of the admissible set and range)**.** *Assume A1–A4. Suppose*

$$\max_{m \in \mathscr{M}} \max_{r \in \mathscr{R}} \left| \hat{s}_r(m) - s_r(m) \right| \xrightarrow{p} 0, \qquad \max_{m \in \mathscr{M}} \left| \hat{\theta}(m) - \theta(m) \right| \xrightarrow{p} 0, \tag{20}$$

and suppose separation: the slack margin $\mu(m) := \min_r s_r(m) + \delta$ satisfies $\mu(m) \neq 0$ for all $m \in \mathscr{M}$. Then $\Pr(\hat{\mathscr{M}}^* = \mathscr{M}^*) \to 1$ and $[\hat{\theta}_L, \hat{\theta}_U] \xrightarrow{p} [\theta_L, \theta_U]$.

Proof sketch. Let $\eta := \min_m |\mu(m)| > 0$ by separation. Uniform consistency gives $\Pr(\max_{m,r} |\hat{s}_r - s_r| < \eta/2) \to 1$. On that event, $\hat{\mu}(m)$ has the sign of $\mu(m)$ for every m, so $\hat{\mathscr{M}}^* = \mathscr{M}^*$. On the same event the endpoints are extrema of $\hat{\theta}$ over the fixed set $\mathscr{M}^*$, and $|\min_{m \in \mathscr{M}^*} \hat{\theta}(m) - \min_{m \in \mathscr{M}^*} \theta(m)| \leq \max_m |\hat{\theta}(m) - \theta(m)| \to_p 0$ (similarly for the maximum). Boundary measures with $\mu(m) = 0$ are the only obstruction: their admission is decided by sampling noise. □

Remark A.1 (Empty sets)**.** *$\mathscr{M}^* = \emptyset$ rejects the pair $(\mathscr{M}, \mathscr{R})$, not the construct C. It is a finding, not an error.*

Remark A.2 (Consistency is not validity)**.** *Proposition A.1 says the plug-in objects are well-posed. Validity claims come from the restrictions themselves (A2) and from the informativeness of Θ^*.*

A.4 Holdout null and claim

Assumption A.5 (Pre-data discipline)**.** *The partition $(\mathscr{R}_S, \mathscr{R}_H)$, tolerance δ, baseline rule B, and random seed are fixed before any holdout data are examined.*

Proposition A.2 (Null and claim)**.** *Under pre-data discipline, $\mathscr{M}_S^*$ does not depend on $\mathscr{R}_H$. Let $H(m) = \mathbf{1}\{m \text{ passes every } r \in \mathscr{R}_H\}$. The claim is $\mathbb{E}[H(m) \mid m \in \mathscr{M}_S^*] > \mathbb{E}[H(m) \mid m \in \mathscr{M}_B^*]$; the null is equality (selection restrictions carry no predictive content for holdout restrictions).*

The natural estimator is the share of survivors passing holdout, $\hat{p} = |\mathscr{M}_S^*|^{-1} \sum_{m \in \mathscr{M}_S^*} H(m)$, compared to the baseline analogue on the same holdout set.

A.5 Coupling and the uncertainty band

The plug-in range inherits sampling error from both $\hat{\theta}(m)$ and $\hat{\mathscr{M}}^*$. When a measure is near the boundary and β-extreme, the two errors are coupled—the difficulty flagged in the NBER program's treatment of inference under measurement menus [14]. Endpoint inference for fixed identified sets [24, 41] takes selection as given; here selection is estimated. The main-text protocol (uncertainty band, empty rate, boundary attribution, optional conservative projection) is an uncertainty summary under a stated reading, not a simultaneous confidence region under arbitrary dependence. Sample splitting decorrelates admission and endpoints at the cost of effective sample size. Formal simultaneous-coverage theory is the subject of the companion methods paper.

A.6 Derivation of DGP A population numbers

Under Table 1 with standardized factors and $\mathrm{Corr}(V_c, C) = 0.80$, $\mathrm{Corr}(Y, C) = 0.60$, $\mathrm{Corr}(V_d, C) = 0$:

$$\begin{aligned}
\mathrm{Corr}(m_{\text{true}}, V_c) &= 0.90 \times 0.80 = 0.72, \\
\mathrm{Corr}(m_{\text{weak}}, V_c) &= 0.50 \times 0.80 = 0.40, \\
\mathrm{Corr}(m_{\text{contam}}, V_c) &= 0.70 \times 0.80 = 0.56, \\
\mathrm{Corr}(m_{\text{method}}, V_c) &= 0.20 \times 0.80 = 0.16, \\
\mathrm{Corr}(m_{\text{noise}}, V_c) &= 0.
\end{aligned}$$

With $\tau_{\text{conv}} = 0.39$, the convergent slacks are $+0.33$, $+0.01$, $+0.17$, -0.23, -0.39 as in Table 2. Discriminant: $\mathrm{Corr}(m_{\text{contam}}, V_d) = 0.55$, so $s_{\text{disc}} = 0.25 - 0.55 = -0.30$; all other menu members have $\mathrm{Corr}(\cdot, V_d) = 0$ and slack $+0.25$. Known-groups: if the latent gap is 0.80, measure gaps are 0.80 times the C-loading, yielding 0.72, 0.40, 0.56, 0.16, 0 and slacks relative to $\tau_{\text{grp}} = 0.25$ of $+0.47$, $+0.15$, $+0.31$, -0.09, -0.25. Targets: $\mathrm{Corr}(m, Y) = \text{loading}_{m,C} \times 0.60$, hence 0.54, 0.30, 0.42, 0.12, 0. These closed forms are the sole source of Table 2; no simulation is involved.

B. Moment-anchored preference policies

Preference optimization from pairwise comparisons identifies, under Bradley–Terry, reward differences relative to a reference policy [6, 37]. When training pairs are generated from *aggregate* population moments z_c (e.g., a country's GPS profile) rather than from individual item responses, the identified object is the set of choice distributions consistent with those moments—again a partial-identification statement in the spirit of (11). What is not identified is cardinal utility, within-country heterogeneity, or cross-country comparability beyond the anchor scale. Item-level pipelines [35] and rationale-conditioned pipelines [34] carry different content assumptions into training; moment anchoring trades those for a transparent aggregation assumption that the validity framework can screen. The trained policy enters $\mathscr{M}$ as one candidate m, not as a privileged structural truth.

C. Notation at a glance

Symbol	Meaning
C	Scientific construct
m	Candidate measure / measurement function
$\mathscr{M}$	Declared menu
V	Auxiliaries used in validity tests
$\mathscr{R}(C)$	Validity tests / nomological network
τ_r	Substantive threshold (researcher-owned)
$\mathscr{R}_S, \mathscr{R}_H$	Selection and holdout partitions
$s_r(m)$	Sample slack of restriction r on measure m
δ	Slack tolerance (default 0)
$\mathscr{M}^*$	Surviving / admissible set
$\mathscr{M}^*_{\text{select}}, \mathscr{M}^*_{\text{robust}}$	Select and robust sets under holdout designs
$\theta(m)$	Scientific result using measure m
Θ^*	Robust result set
$[\theta_L, \theta_U]$	Construct-identified range
$G(D, C)$	Measurement-generation technology

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